



SINATRA CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & BASIC APP

Installation

```
gem install sinatra
# app.rb
require 'sinatra'
get '/' do
  'Hello World!'
end
ruby app.rb # starts on port 4567
rerun app.rb # auto-reload on change
```

04. VIEWS

Workflow

```
# views/ directory with .erb files
get '/users' do
  @users = User.all
  erb :users_index # renders views/users_index...
end
get '/page' do
  haml :page # or slim, liquid, etc
end
# Inline:
erb '<%= @name %>', locals: { name: 'Alice' }
```

02. ROUTES

Architecture

```
get '/users' do; json User.all; end
post '/users' do
  user = User.create(JSON.parse(request.body.re...
  status 201; json user
end
put '/users/:id' do; ... end
delete '/users/:id' do; User.find(params[:id])...
```

05. BEFORE/AFTER FILTERS

CRUD API

```
before do
  @current_user = User.find_by(token: request.e...
  halt 401, json(error: 'Unauthorized') unless ...
end
before '/admin/*' do
  halt 403 unless @current_user.admin?
end
after do
  headers 'Content-Type' => 'application/json'
end
```

03. PARAMS

YAML / Config

```
get '/users/:id' do
  id = params[:id] # route parameter
  User.find(id).to_json
end
get '/search' do
  q = params[:q] # query string ?q=alice
  User.where(name: q).to_json
end
# POST body (form-encoded):
name = params[:name]
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. SESSIONS

Hardening

```
enable :sessions
set :session_secret, 'change-me-in-production'
# In route:
session[:user_id] = user.id
user_id = session[:user_id]
session.clear
# Cookie-based by default; use rack-session-p...
```

10. COMMON ROUTE HELPERS

Unit Tests

```
json(data)           # {Content-Type: appl...
status 201           # set response status
headers 'X-Foo' => '1' # set response headers
send_file 'file.pdf'  # file download
stream { |out| out << 'chunk' } # streaming r...
request.body.read     # raw request body
request.ip            # client IP address
```

08. HALT & REDIRECT

Observability

```
halt 404, 'Not found'
halt 401, json(error: 'Unauthorized')
redirect '/'
redirect '/login', 302
redirect back # to referrer
# In before filter:
halt(401, {error: 'No token'}.to_json) unless...
```

11. COMMON GOTCHAS

Commands

Classic mode pollutes global scope use Modular for production
enable :sessions stores in cookie set strong secret
Sinatra halts after first matching route
request.body.read can only be read once rewind if needed
No built-in ORM pair with Sequel, ActiveRecord, or ROM

09. MODULAR vs CLASSIC

Optimization

```
# Classic (top-level DSL, one app):
require 'sinatra'
get '/' do; 'Hello'; end
# Modular (multiple apps, explicit run):
require 'sinatra/base'
class MyApp < Sinatra::Base
  get '/' do; 'Hello'; end
end
# config.ru: run MyApp
```